The left probability range from five neighboring counts

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Five neighboring counts retain enough of the crossing loss to pay for the finite corrections. Centered replacements compare their smooth kernels without an inward-displacement error. On the left probability range this gives a complete signed exclusion at every index at least 700, including the noise between the previous small-noise theorem and the geometric ceiling.

Write $\phi_0 = (2\pi)^{-1/2}$ and $C_E = (3+\sqrt{10})\phi_0/6$.

Theorem 1. *Let $n \ge 700$ be an integer, $p \in [1/3, 2/5]$, $q = 1-p$, and $Y = B + U$, where B is Bernoulli of parameter p and $\mathbb{E}(U \mid B) = 0$. Write U_j for the conditional noise given $B = j$. Suppose*

$$-7/10 \le U_0 \le a_0, \quad -a_1 \le U_1 \le 7/10, \quad a_0, a_1 \ge 0, \quad a_0 + a_1 \le 17/250.$$

Put $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (Y-p)/\sqrt{v+s}$ and $\rho = \mathbb{E}|X|^3$. If $1/100 \le \lambda \le 8/25$ and $|H| \le 46/25$, then for independent copies $X_1, \dots, X_n$,

$$\sqrt{n}\{\mathbb{P}(X_1 + \cdots + X_n \le H) - \Phi(H/\sqrt{n})\} - C_E\rho < -\lambda/100.$$

There is no contact, predecessor or geometric-budget premise in this comparison theorem. Those premises remain in acquiring the displayed strip and noise ceiling for a putative extremal witness. The conclusion is signed; reflection requires the inputs for the reflected law.

1 Retaining the exact crossing loss

Set $t = np + H\sqrt{v+s}$, $k = \lfloor t \rfloor$, $u = t - k \in [0,1)$, $\delta = k - np$, and $s_j = \mathbb{E}U_j^2$. Since $v \in [2/9, 6/25]$,

$$v + s < 1/4, \quad |t - np| < 23/25, \quad -48/25 < \delta < 23/25.$$

In particular all five indices $k-1, \dots, k+3$ are interior. Use independent conditional pairs to form

$$T_j = j + \sum_{i \le j} U_{1i} + \sum_{i > j} U_{0i}, \quad V_i = \sum_{l < i} U_{1l} + \sum_{l > i} U_{0l},$$

$$h(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \le x), \qquad g_i = \mathbb{E}h(t - i + 1 - V_i).$$

The mesh is increasing; $T_0 \le t < T_n$ follows from the inward caps. The finite crossing identity of v13 gives $g_i \ge 0$ and, with $\pi_j = \mathbb{P}(\mathrm{Bin}(n,p) = j)$,

$$\mathbb{P}(X_1 + \cdots + X_n \le H) = (1-u)F_B(k) + uF_B(k+1) + \mathcal{C}(t) - \sum_i \pi_i g_i, \tag{1}$$

where

$$\mathcal{C}(t) = \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j)\{\mathbb{E}(t - T_j)_+ - (t-j)_+\}.$$

Each conditional sum has the distribution from the actual iid law. The auxiliary coupling changes none of these conditional marginals. Lower crossings remain closed and upper crossings strict, including atoms.

2 Five smooth kernels give the reserve

Let $B(x) = \sin^2(\pi x)$ on $[0,1]$ and zero elsewhere. Its derivative is globally Lipschitz with constant $2\pi^2$ and is zero at both joins. We first check the pointwise minorant

$$h(x) \geq (2/5)B(x). \tag{2}$$

Separated support gives $h \geq 0$. If $0 \leq x < 1 - a_1$, Jensen gives $h(x) \geq x$. The elementary bound $B(x) \leq (5/2)x$ follows from $\pi^2 < 10$ for $x \leq 1/4$, the tangent at $1/4$ on $[1/4, 2/5]$, and $B \leq 1$ for $x \geq 2/5$. On the middle interval B is concave and its tangent is $1/2 + \pi(x - 1/4)$; the two endpoint comparisons use $\pi < 10/3$.

For $1-a_1 \leq x \leq 1$, put $D = 1+U_1$ and $L = \mathbb{E}(D-x)_+$. Put–call parity gives $h = \mathbb{P}(D > x) - L$, and hence

$$h(x) \geq \frac{x - 7/10}{17/10 - x} L \geq \frac{29}{96}(1 - x),$$

since $x \geq 233/250$ and $L \geq 1 - x$. On this same interval,

$$(2/5)B(x) \leq (2/5)(22/7)^2(17/250)(1 - x) < (29/96)(1 - x).$$

At $x = 1$ the minorant is zero. The strict survival probability in this calculation retains all atoms of D.

Freeze $V = V_{k+1}$ and compare the five omitted-coordinate laws $V^{(j)} \overset{\text{law}}{=} V_{k+1+j}$, $-2 \leq j \leq 2$. If F' is Lipschitz with constant L and Z is centered, Taylor's integral remainder gives $|\mathbb{E}F(r + Z) - F(r)| \leq L\mathbb{E}Z^2/2$. Thus one centered replacement costs at most $\pi^2(s_0 + s_1)$ for a bump with this derivative bound. There are only four replacements after regrouping: for example, with $B_j(x) = B(u - j - x)$, the two positive-index differences equal

$$\begin{aligned}\mathbb{E}(B_1 + B_2)(V^{(1)}) - \mathbb{E}(B_1 + B_2)(V) \\ + \mathbb{E}B_2(V^{(2)}) - \mathbb{E}B_2(V^{(1)}).\end{aligned}$$

The open supports of B_1, B_2 are disjoint, and their derivatives vanish at the common boundary. Their sum still has derivative Lipschitz constant $2\pi^2$. The negative indices give two analogous replacements. Since $s_0 + s_1 \leq 3\lambda/n$, their total cost is at most $12\pi^2\lambda/n$. No independence of sums at different counts is assumed.

The infinite translates of B sum to a periodic sine square. Every omitted nonzero term with $|j| \geq 3$ requires $|V| > 2$, so

$$\sum_{j=-2}^{2} \mathbb{E}B(u - j - V) \geq \mathbb{E}\sin^2(\pi(u - V)) - \mathbb{E}V^4/16.$$

The two boundary bumps vanish at integer joins, so there is no missing atom contribution.

For $|x| \leq 96/125$, one has

$$1 - \cos(2\pi x) \geq (7/5)x^2. \tag{3}$$

On $[0, 3/4]$, the sine chord gives $\sin(\pi x) \geq (2\sqrt{2}/3)x > (14/15)x$. On $[3/4, 96/125]$, its derivative bound gives $\sin(\pi x) > 7/10 - (22/7)(9/500) = 563/875$. Squaring these positive bounds proves (3) on both closed intervals, then evenness completes the proof. Conditional support diameters are at most $7/10 + 17/250 = 96/125$. For either conditional noise Z and an independent copy Z', this yields

$$1 - |\mathbb{E}e^{2\pi i Z}|^2 \geq (14/5)\operatorname{Var} Z, \qquad |\mathbb{E}e^{2\pi i Z}| \leq e^{-(7/5)\operatorname{Var} Z}.$$

Only the independent frozen summands are multiplied. Their variance is

$$\tau = \mathbb{E}V^2 = (nq - \delta - 1)s_0 + (np + \delta)s_1, \qquad (1 - 6/n)\lambda \leq \tau \leq (1 + 3/n)\lambda.$$

These are the physical count coefficients; they follow from the displayed bounds for δ, $p \ge 1/3$ and $q \ge 3/5$. The fourth moment is at most $(49/100)\tau + 3\tau^2$. Therefore

$$\sum_{i=k-1}^{k+3} g_i \ge \frac{2}{5}\left\{\frac{1-e^{-(7/5)(1-6/n)\lambda}}{2} - \frac{(49/100)(1+3/n)\lambda + 3(1+3/n)^2\lambda^2}{16} - \frac{12\pi^2\lambda}{n}\right\}. \tag{4}$$

This is the left-strip adaptation of the four-replacement proof in `loss-direct.tex`; the changes of support width, probabilities and variance constants have all been derived here.

3 Binomial weights and finite corrections

For $|i-k-1| \le 2$, the phase bounds give $|i-np| < 4$. The segment from i/n to p has variance product greater than $11/50$. Finite Stirling with remainder $0 < r_j < 1/(12j)$ and the entropy Taylor bound imply, uniformly for $n \ge 700$,

$$\sqrt{n}\pi_i \ge 2\phi_0 \exp\left\{-\frac{8+1/12}{700(11/50)}\right\} > 2\frac{19943}{50000}\left(1 - \frac{8+1/12}{700(11/50)}\right) > \frac{3}{4}.$$

Thus (4) can be multiplied by $3/4$. The other count losses are nonnegative and need no tail estimate.

Here is the summable curvature bound from v14, in its full form. If $c = (np-q-t)_+ \le 1$, T bounds the variances of the increasing-mass counts, M bounds their fourth moments, and $\max_r \mathbb{P}(\mathrm{Bin}(n+1,p) = r) \le \beta/\sqrt{(n+1)v}$, then

$$\sqrt{n}\mathcal{C}(t) \le \frac{\beta}{nv^{3/2}}\left\{T/4 + c\sqrt{T} + \frac{27M}{256}(5/3 + 5c/4)\right\}. \tag{5}$$

Indeed the adjacent binomial difference has numerator $np-q-j$. For $a = t-j$, the centered stop-loss bend is bounded by $\sqrt{T}/2$, by $T/(4a)$ when $a > 0$, and by $27M/(256|a|^3)$ away from zero. At most two nearby points are treated directly; the remaining full sums use $\sum i^{-2} < 5/3$ and $\sum i^{-3} < 5/4$. The increasing-mass indices satisfy $j < np$, so we may take

$$T = 5\lambda/3, \quad M = (49/100)T + 3T^2, \quad \beta = 41/100.$$

For completeness the last bound follows by finite Stirling at a mode: for $N = n+1 \ge 43$, its variance product exceeds $39v/40$, whence the mass is smaller than $(2/5)\sqrt{40/39}(516/515)/\sqrt{Nv} < (41/100)/\sqrt{Nv}$. Also $v^{3/2} > 13/125$, and

$$\sqrt{T} \le (250/39)\lambda + 13/200$$

follows by squaring $\sqrt{T} - 13/100$.

Schulz's strict finite-n Bernoulli theorem gives at both integer anchors $\sqrt{n}\{F_B(j) - \Phi((j-np)/\sqrt{nv})\} < C_E\rho_B$, where $\rho_B = (1-2v)/\sqrt{v}$. Only this all-p theorem is needed, not its sharper restricted-probability version. Linear interpolation preserves the strict inequality. Put $w = t - np$. Its normal interpolation error is at most

$$\frac{\phi_0}{2nv^{3/2}}u(1-u)(u-w)_+.$$

This follows by bounding the positive part of the normal CDF's second derivative over the whole unit interval. For $H \le 0$, $c < 8/25$, the variance change is favorable and the remaining normal error is at most

$$\frac{2/5}{nv^{3/2}}(23/200 + 2/27).$$

For $H \ge 0$, $c = 0$, and interpolation plus variance repair is at most

$$\frac{2/5}{nv^{3/2}}(2/27 + (23/50)\lambda).$$

These use $|w| < 23/25$, $u(1-u) \le 1/4$, $u^2(1-u) \le 4/27$ and $1 - (1+x)^{-1/2} \le x/2$.

The actual cubic correction is favorable:

$$(v+s)^{3/2}(\rho - \rho_B) \ge s\{1 - 5/6 - 17/250 - (15/16)s\} > 0,$$

since $s \le (8/25)/700$. To obtain this bound, expand the marginal absolute third moment and retain its signs, use $\mathbb{E}U_0^3 \le a_0 s_0$, $\mathbb{E}U_1^3 \ge -a_1 s_1$, and $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$. No favorable cubic sign is inferred below $p = 1/3$.

4 Two concave quartics close the entire noise interval

Set $N = 700$, $a = 1 - 6/N$, $b = 1 + 3/N$, $z = (7/5)a$ and define

$$W(\lambda) = \frac{3}{10}\left\{\frac{z\lambda - z^2\lambda^2/2 + z^3\lambda^3/6 - z^4\lambda^4/24}{2} - \frac{(49/100)b\lambda + 3b^2\lambda^2}{16} - \frac{12(22/7)^2\lambda}{N}\right\}.$$

Taylor's theorem gives $1 - e^{-x} \ge x - x^2/2 + x^3/6 - x^4/24$ for $x \ge 0$. Every term in (4) improves with n, so W is a lower bound for the normalized weighted loss at every $n \ge N$. Write

$$S_c = T/4 + c\{(250/39)\lambda + 13/200\} + (27M/256)(5/3 + 5c/4),$$

$$E_-(\lambda) = \frac{(41/100)S_{8/25} + (2/5)(23/200 + 2/27)}{13/125},$$

$$E_+(\lambda) = \frac{(41/100)S_0 + (2/5)(2/27 + (23/50)\lambda)}{13/125}.$$

These are nonnegative quadratics. The preceding bounds reduce the theorem to $P_\pm(\lambda) > 0$ on $[1/100, 8/25]$, where

$$P_\pm = W - E_\pm/N - \lambda/100.$$

The gain polynomial is strictly concave on the whole real line, since

$$W''(\lambda) = -\frac{3}{10}\left\{\frac{z^2}{4}\big((z\lambda - 1)^2 + 1\big) + \frac{3b^2}{8}\right\} < 0.$$

Both error quadratics are convex. Hence $P_\pm'' < 0$, and their minima on the closed noise interval occur at endpoints. The four exact endpoint margins are positive. The smallest is

$$\frac{5442273179742877948871}{97819312500000000000000000} > 0.$$

The companion rational checker reconstructs all coefficients and guards; it does not accept sampled positivity or a fitted grid. This proves Theorem 1 with all phases and noise endpoints included.

Scope and evidence

The kernel and four-replacement method are in this round's `loss-direct.tex`; the v13 crossing identity and v14 summable-curvature proof retain their stated atom conventions. The external Bernoulli theorem is Schulz, *The optimal Berry–Esseen constant in the binomial case* (2016), Theorem 1, printed page 1, original thesis. Finite Stirling and the normal constants are explicitly enclosed in the checker. No Fourier or certificate-generation campaign is used.

Combined with the retained LEFT-01 comparison below $\lambda = .01$, this closes the acquired $p \in [1/3, .4]$ strip at all $n \ge 700$ through its noise ceiling. The retained geometry excludes $p < 1/3$ already at $n \ge 169$. Acquisition, reflection, attainment, remaining finite indices and BASE41 remain separate obligations. No unconditional cutoff theorem is claimed.